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Experimental study on suppressing vortex-induced vibration of a long-span bridge by installing the wavy railings

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ABSTRACT

It is known that vortex shedding behind bluff bodies can be suppressed by introducing some three-dimensional geometric disturbance. Therefore, a new passive control device named wavy railings is proposed to mitigate the vortex-induced-vibration (VIV) of a single box section bridge. The wavy railing is composed of vertical posts and wavy crossbars, which can provide three-dimensional perturbation to the flow field and then lead to the development of a three-dimensional free shear layer which does not roll up into a vortex street and hence suppress the VIV. In this study, the control effects of the wavy railings on mitigating VIV are investigated through wind tunnel tests. The results reveal that when the wavelength of the wavy bar is 0.5H or 1H (H is the height of the bridge deck), both vertical and torsional VIV can be completely suppressed, and the upper surface edge is the optimal installation position.

1. Introduction

With the increase of bridge span, modern bridge structures become more flexible, slender, less damping, and therefore more sensitive to the wind effects. Vortex-induced vibration (VIV), which belongs to self-limited amplitude vibration, is a typical type of wind-induced vibrations in long span bridges. Although the VIV generally does not lead to catastrophic results, it can cause large amplitude vibration within a relatively low wind speed range, which inevitably brings discomfort to drivers and causes structural fatigue damage to a certain extent. Therefore, reducing VIV amplitudes or even completely eliminating VIV has become one of the essential requirements in the design stage of modern long-span bridges (Hu et al., 2019).

As the main source of VIV, the vortex shedding is principally governed by the geometric configuration of the bridge section (Xu et al., 2016). Therefore, the selection of main girder section plays a crucial role in the VIV performance. Generally, streamlined design is benefit for the bridge girder to achieve a better aerodynamic performance. However, for bridges in service, it is necessary to install some indispensable attachments for safety and serviceability, such as railings, median dividers and maintenance rails. These attachments dramatically increase the overall degree of bluffness of the bridge deck section (Bruno and Mancini, 2002), and thus deteriorate the aerodynamic performance of the streamlined girder to some extent. As reported in Laima et al. (2018), because the bridge attachments are very close to the bridge deck surface, the bridge deck boundary layer is severely disturbed, resulting in significant effects on the aerodynamic characteristics of the bridge deck. In recent decays, there have been quantities of efforts devoted to studying the aerodynamic effects of various bridge attachments. Nagao et al. (1997) experimentally found that the size and position of the handrail's horizontal bar significantly influenced the flow separation and the torsional VIV of the bridge deck. Guan et al. (2014) concluded that the vertical VIV of a bluff box girder can be suppressed by changing the wind resistant ratio of the railings. As discovered by Haque et al. (2016), the aerodynamic response generated by the perforated handrails was lower than that of the solid one due to the reduced total wake size. Similar findings were also reported in Wang et al. (2011). Zhu et al. (2015) investigated the influence of bridge attachments on the VIV performance of a streamlined single-box girder. In their study, the type of railings on the sideway significantly affected the vertical and torsional VIV and the maintenance rail on the leeward side was the key factor to induce the VIV. Fang et al. (2017) studied a semi-open separated twin-box deck which was subjected to significant vertical vortex-induced vibration at the attack angle of $+3°$ or $+5°$. The results indicated that the flow

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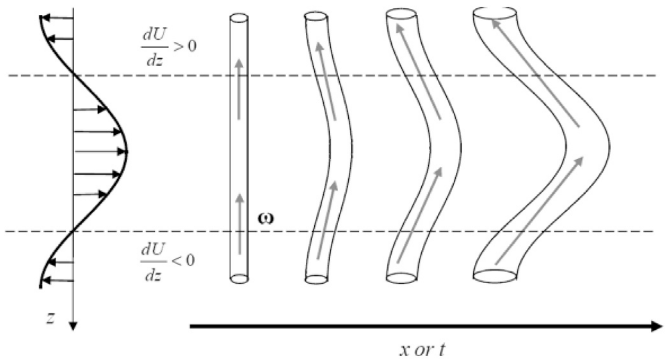


Fig. 1. Evolution of spanwise vortex with three-dimensional controls (Hwang et al., 2013).

separation generated at the railings on the side way was the dominant cause of VIV, and the vibration amplitude could be amplified by maintenance rails. Both Xian et al. (2009) and Li et al. (2011) indicated that the amplitude of the VIV can be reduced by optimizing the location of maintenance rails. Xin et al. (2018a) compared the VIV response of a closed box bridge deck with and without railings through the wind tunnel tests, and observed that the addition of the railings increased the VIV response peak and widened the lock-in range. As reported in Larsen and Wall (2012), the presence of the railings substantially reinforce the vortex shedding response, resulting in the vortex shedding frequency jumps from higher frequencies to lock-in with the bending frequency. In conclusion, these findings indicated that the bridge attachments have great impact on the VIV performance of bridge, and in most cases the impact is negative.

Nowadays, many forms of control methods have been proposed for suppressing VIV of bridges, among which passive aerodynamic countermeasures are usually preferred because of their relative reliability, simplicity and environmental protection. Passive aerodynamic methods are realized by adding passive control devices which do not consume external energy or optimizing the aerodynamic shape of the bridge girder. They are broadly classified into two-dimensional (2D) methods and three-dimensional (3D) methods. In terms of 2D methods, the underlying mechanism to improve the aerodynamic characteristic is to stabilize the wake instability by modifying the 2D section shape of the main girder or by adding various 2D wind-resistance devices whose section shape remains constant along the span of the bridge. Conventional 2D control devices are usually attached to the leading and trailing edges of the deck, such as guide vanes, fairing, flaps, wind spoiler and deflector (Fujino and Siringoringo, 2013). Among them, guide vanes are more widely used due to their prominent control effect. They were installed on the Great Belt East suspension bridge and successfully suppressed the vortex-induced vibration (VIV) (Larsen et al., 2000). On the other hand, 3D control method has recently been considered as a very efficient measure (Kim and Choi, 2005). The periodic spanwise perturbations provided by the 3D controls strongly affect the near-wake region, leading to the development of the three-dimensional shear layers making them less susceptible to rolling-up into a Karman vortex street and hence suppressing the VIV. Hwang et al. (2013) have reported that the formation of streamwise vortices from tilting of two-dimensional Karman vortices (Fig. 1) and the subsequent tilting of these streamwise vortices by the spanwise shear in the base flow are the two key processes to regulate the vortex shedding. And the optimal spanwise perturbation may trigger a resonant response with the mode-A instability (Barkley and Henderson, 1996; Williamson, 1996a, b; Wu et al., 1996), which can significantly distort the original 2D base flow and enhance the 3D characteristics of wake vortex structures. Similar results were obtained in the research studies of Lasheras and Choi (1988), Meiburg and Lasheras (1988), Lasheras and Meiburg (1990), Darekar and Sherwin (2001), Doddipatla (2008), and Xin et al. (2018b). They also observed that the

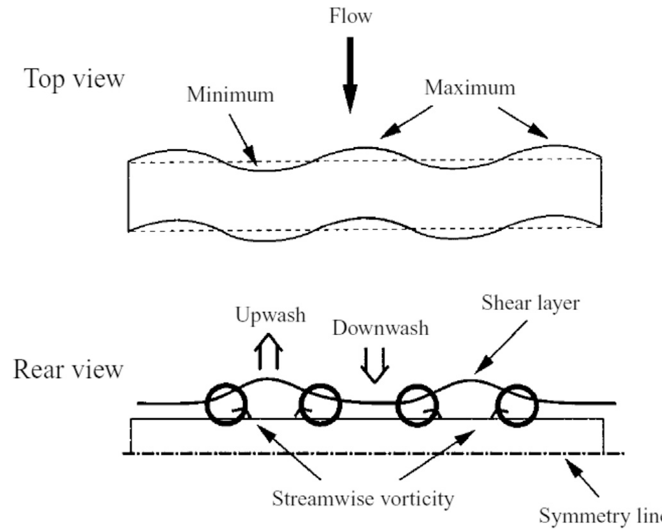


Fig. 2. Schematic view of the relation between the spanwise vorticity and wake width variation along the span (Darekar and Sherwin, 2001).

spanwise disturbance provided by many 3D controls resulted in the formation of counter-rotating pairs of streamwise vortex tubes that were located in the braids, connecting consecutive Karman vortices of opposite sign. In addition, there was a presence of streamwise and vertical vorticity within the shear layers which did not exist in the base flow. This three-dimensional redistribution of vorticity may prevent the shear layers interacting in the near-base region and hence suppress the vortex shedding. In recent years, the application of passive 3D methods to the bridge field has attracted much attention. Xin et al. (2018a) successfully suppressed both the vertical and torsional VIV of a single box girder bridge by setting passive vortex generators at the trailing edge of the bottom surface, which could directly generate streamwise vortex to the wake thus resulting in the secondary wake instability mode. Chen et al. (2019) applied the passive-jet method to a bridge deck and validated its control effect of completely suppressing the vertical VIV and reducing the maximal amplitude of torsional VIV by 45.5%. It was noted that the control effect of this approach was sensitive to the location of the jet slits and the upper part of the bottom inclined panel was the optimal choice. In addition to the above strategies, spanwise periodic perturbation method (SPPM) is another efficient passive 3D control method, which applies periodic geometric distortions to the leading or trailing edge or the face of the body in question and can result in the suppression of the vortex shedding and a reduction of drag force (Dobre et al., 2006). The wavy bluff with surface profile varying sinusoidally along the spanwise direction is a classic application of the SPPM and the related studies were summarized in Lam and Lin (2009). Bearman and Owen (1998) applied the spanwise waviness to thin plates and rectangular cross-section bodies. They found that the introduction of a spanwise wave with a wave steepness in excess of between 0.06 and 0.09 could result in the complete suppression of vortex shedding and substantial drag reduction of at least 30%. Darekar and Sherwin (2001) numerically investigated the flow past a wavy square cylinder at low Reynolds numbers and found that the formation of the 3D near wake was due to the fact that the streamwise vorticity created an upwash behind the maximum and a downwash behind the minimum as shown in Fig. 2. El-Gammal et al. (2007) applied this method to control the vertical VIV of a plate girder bridge and verified the control effect through the wind tunnel test. In the test, the shape of the leading and trailing edges of the girder was modified to sinusoidal waveforms which could provide three-dimensional geometric perturbation to the wake. However, in practical engineering, this form of girders may increase the prefabrication cost and construction difficulty. Inspired by the theory of SPPM and considering that the close relationship between the bridge VIV performance and attachments, we

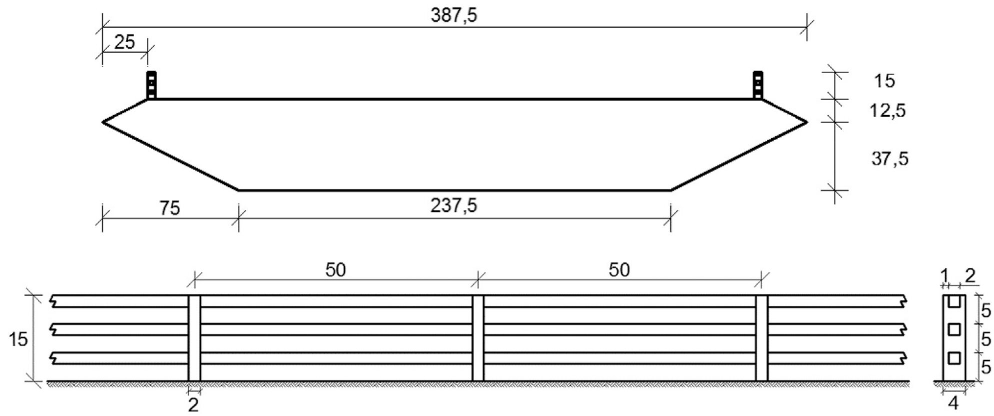


Fig. 3. Geometrical sizes of (a) the test model (b) the base railing (unit: mm).

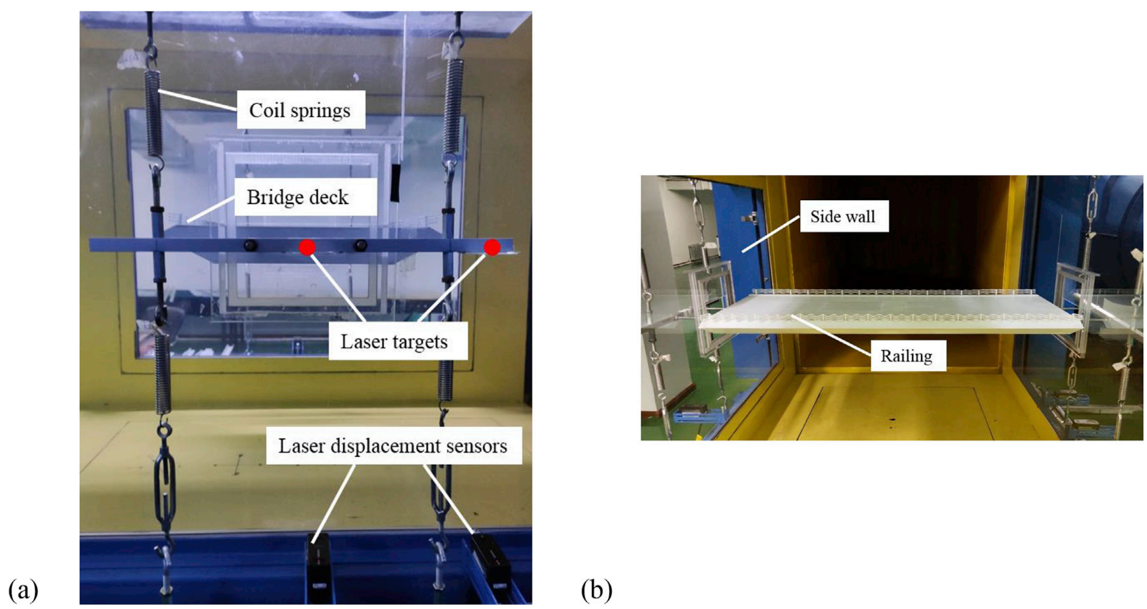


Fig. 4. (a) The side view and (b) the front view of the suspension system.

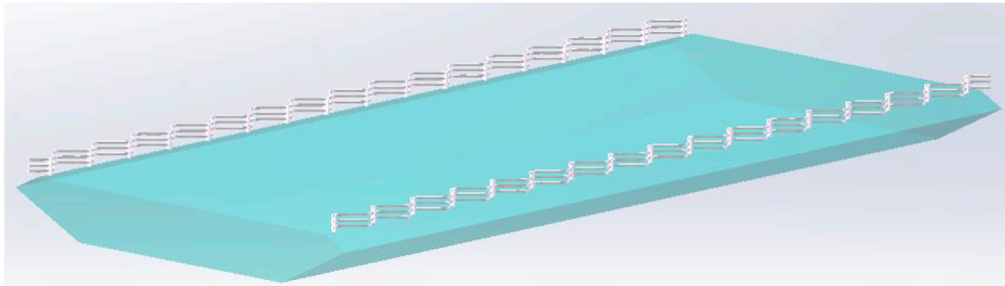
designed a new type of railing with sinusoidal wavy crossbars, named wavy railing, to mitigate the responses of a bridge which suffered VIV without control. It can be seen that wavy railings can not only transform the potential negative impact of normal railings on the aerodynamic performance into a positive impact, but also replace the pure wind resistance devices that cannot provide service function for bridge users.

In this paper, a parametric study for the control effects of the wavy railings on the VIV of a single box girder bridge was conducted in detail. The objectives are 1) to verify the effectiveness of the proposed wavy railing on controlling VIV 2) to investigate the influence of different control parameters to find the optimal configuration of the wavy railings 3) to compare the wake characteristics of the bridge with traditional railings and wavy railings to further explore the control mechanism.

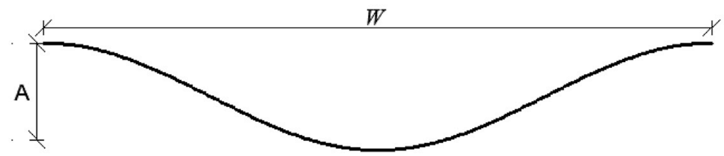
2. Experimental set-up

2.1. Models

The prototype of the sectional bridge model used in this study is the Great Belt East Bridge, which suffered from a severe VIV within the wind speed range of 4–12 m/s (Larsen and Wall, 2012). The sectional bridge deck model with 1:80 scale ratio was 0.83 m long, had the breadth (B) of 0.3875 m and height (*H*) of 0.05 m. Taking into account the height of railings, the aspect ratio of the model was roughly 6:1 as shown in Fig. 3 (a). The surface of the girder was made of acrylonitrile-butadiene-styrene (ABS) with thickness of 3 mm and stiffening ribs were rigidly installed inside the model to insure the adequate stiffness. The base railing was strictly designed and manufactured following the principle of geometric similarity, with a height of 0.015 m, consisting of 3 straight horizontal crossbars with a 2 mm × 2 mm square section and 17 vertical posts with a 2 mm × 4 mm rectangular section. The constant spacing of the vertical posts was 50 mm and the crossbars were also equally spaced, the detailed geometric dimensions are depicted in Fig. 3 (b). The railings were made of ABS and produced by a laser engraving machine to ensure the dimension accuracy. They were tightly attached to the surface of the bridge to prevent malposition or vibration during the test. The mass and the mass inertia of the overall model including the bridge deck, railings and the contribution from the system were: $m = 4.319\text{kg/m}$ and $I = 0.053\text{kgm}^2/\text{m}$, yielding dimensionless values $m/\rho B^2 = 23.46$ and $I/\rho B^4 = 1.92$, where $\rho = 1.226\text{kg/m}^3$ is the air density. The model was mounted inside the wind tunnel by using an external suspension system consisting of lever arms and eight coil springs, which permitted both heave and pitch stiffness of the bridge as shown in Fig. 4. The side walls of the wind tunnel acted as the end walls to eliminate the effect of the supporting system and end effects. The natural frequencies and damping ratios of the system were identified by conducting a free decay



(a)



(b)

Fig. 5. (a) Layout and (b) geometric parameters of the wavy crossbar.

test. The vertical and torsional frequencies of the test modal in still air were 8.5 Hz and 12.33 Hz respectively, and the damping ratios were $\zeta_v = 0.24\%$ and $\zeta_\alpha = 0.19\%$.

To enable the railing to possess both the security assurance and wind-resistance function, the straight crossbars of the base railing were modified to the wavy crossbars with sinusoidal waveforms parallel to the deck, the layout of the wavy railing is shown in Fig. 5 (a). Moreover, because the railings should be arranged symmetrically on both sides of the deck, the problem of uncertainty wind direction can be solved directly. The height and windward area of the wavy railing remain the same as the base railing, so the initially solidity ratio remains constant, which is closely related to the static aerodynamic force (Bruno and Mancini, 2002; Wang et al., 2011). Based on the theory of the SPPM, the wavelength *W* and the peak-to-peak amplitude *A* of the sine wave are the key control parameters as depicted in Fig. 5 (b). However, it is not clear what wavelength and amplitude should be selected, because in previous studies, geometric distortion was usually applied to a single blunt body, while in this study, distortion was generated by the attachment of the composite structure. Therefore, on the premise of ensuring the rationality of the railing shape, a detailed parametric study was conducted. The wavelengths (*W*) selected were 0.5*H*, 1*H*, 2*H*, 3*H*, 4*H* and 5*H*, matched with the selected peak-to-peak amplitude (*A*) of 0.25*H*, 0.3*H*, 0.36*H* and 0.4*H*. In addition, to investigate the effect of the wavy railing's position on control effects, we designed 3 different installation positions where the wavy crossbars were completely outside the deck surface, half outside the deck surface and completely inside the deck surface, respectively denoted by *T*, *M* and *L*. The top view of the wavy crossbars at three positions relative to the edge of the deck surface is shown in Fig. 6, in which the black solid line represents the edge of the deck surface. In total, 31 test cases were adopted and a naming standard was developed as shown in Table 1. For example, *T-W*1*H-A*0.25*H* represents that the downward projection of wavy crossbars is located outside the deck surface, with a wavelength of 1*H* and a peak-to-peak amplitude of 0.25*H*, and the same logic to the other names. Besides, 'Straight railing' denotes the unmodified traditional railings.

Table 1
Test cases for control effect of wavy railings with various control parameters.

Case	Wavelength/*H*	Amplitude/*H*	Position of the wave
Straight railing	–	–	–
T-W0.5H-A0.25/0.3/0.36/H	0.5	0.25/0.3/0.36/0.4	Outside the deck surface
T-W1H-A0.25/0.3/0.36/H	1	0.25/0.3/0.36/0.4	Outside the deck surface
T-W2H-A0.25/0.3/0.36/H	2	0.25/0.3/0.36/0.4	Outside the deck surface
T-W3H-A0.25/0.3/0.36/H	3	0.25/0.3/0.36/0.4	Outside the deck surface
T-W4H-A0.25/0.3/0.36/H	4	0.25/0.3/0.36/0.4	Outside the deck surface
T-W5H-A0.25/0.3/0.36/H	5	0.25/0.3/0.36/0.4	Outside the deck surface
M-W1H-A0.36H	1	0.36	Half outside the deck surface
L-W1H-A0.36H	1	0.36	Inside the deck surface
M-W3H-A0.36H	3	0.36	Half outside the deck surface
L-W3H-A0.36H	3	0.36	Inside the deck surface
M-W5H-A0.36H	5	0.36	Half outside the deck surface
L-W5H-A0.36H	5	0.36	Inside the deck surface

2.2. Wind tunnel set-up and test measurements

The present experiment was conducted in a closed-circuit wind tunnel at the Northeast Forest University in China. The test section was 5 m in length, 1.0 m in height, and the width gradually increased from 0.8 m to

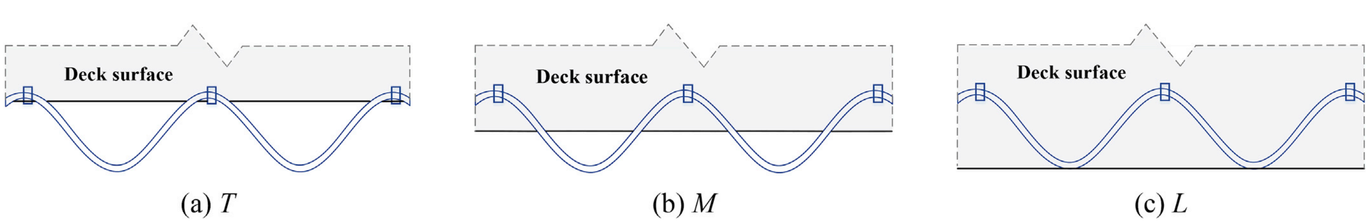
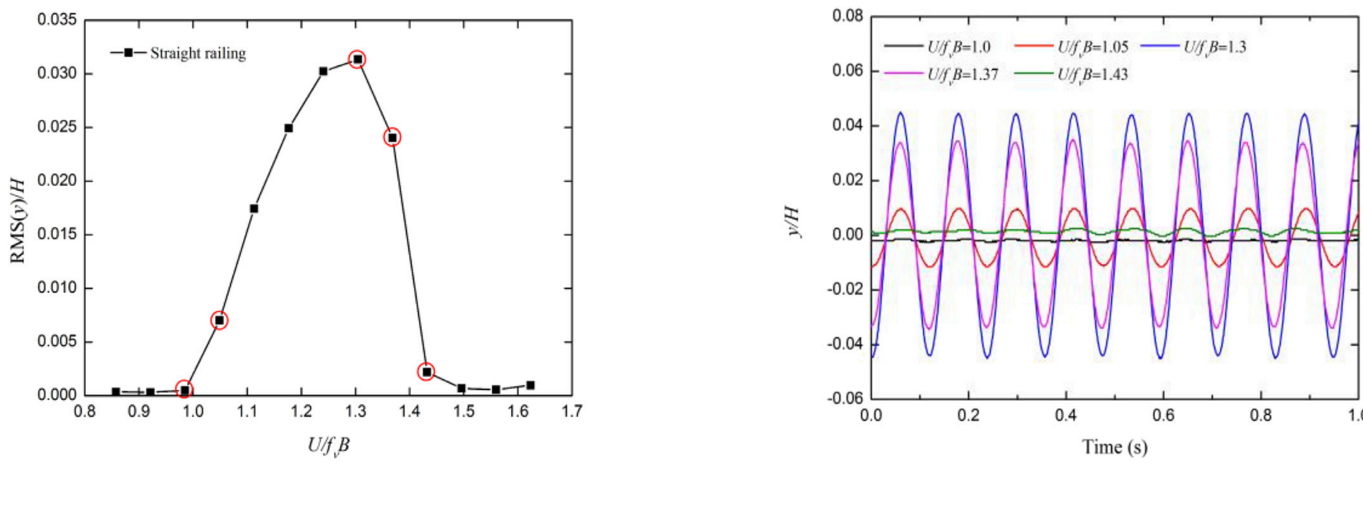


Fig. 6. Top view of wavy railings located in different positions.



(a) (b)

Fig. 7. (a) RMS of the vertical response against the reduced wind velocity, (b) Response time history.

0.9 m along the inflow direction. The top and side walls of the test section were made of glass for convenient observation. Honeycombs and screen structures were installed before test section to improve the quality of the incoming flow. The wind velocity range available was 2 m/s to 70.5 m/s with a minimum interval of 0.1 m/s. The maximum free stream turbulence intensity was 0.5% and the maximum free-stream non-uniformity was 0.5%. In this study, the model was mounted in the last test section along the wind direction and the oncoming flow in the test cases was uniform and smooth. The wind speed range of interest was2.8 − 8.2m/s, corresponding to the Reynolds number range of $0.7\times\ 10-2.1\times\ 10^5$(based on the breadth B).

The wind-induced responses of the sectional model were monitored by two laser reflection sensors (IL-300, Keyence, Japan), mounted in pairs on the side of the support system and aimed at reflected targets set on the lever arms (see Fig. 4 (a)). The time history signals from the reflection sensors were acquired by using a National Instruments (NI) acquisition system (PXIe-1082, National Instruments Corporation, USA) at a sampling frequency of 1000 Hz for 60s. In the present calculations, the vertical and torsional displacements of the model were defined by

$$y=(y_1+y_2)/2$$

$$\theta=180\times(y_1-y_2)/\pi d$$

where y and θ are the vertical and torsional displacement, respectively; y_1 and y_2 are the vertical displacement of two measurement spots on the lever arms; d is the horizontal distance between the two measurement spots, which are marked in Fig. 4 (a). (Yuan et al., 2017) It's noted that all the measured data were acquired after a waiting interval of 120s when the vibration system was in the stable state.

3. VIV response

In this section, we first focus attention on studying the VIV performance of the bridge model with the traditional railings, which was regarded as the base case. Then, a series of vibration tests was carried out to verify the control effect of the wavy railings on VIVs, and the influence of different control parameters, i.e. the wavelength, amplitude and position of the wavy crossbars was investigated in detail through contrastive analysis. The test results are presented as plots of vertical VIV (V-VIV) and torsional VIV (T-VIV) response as function of the non-dimensional wind speed $U/f_\nu B$ and $U/f_\alpha B$, where $U$is the incoming

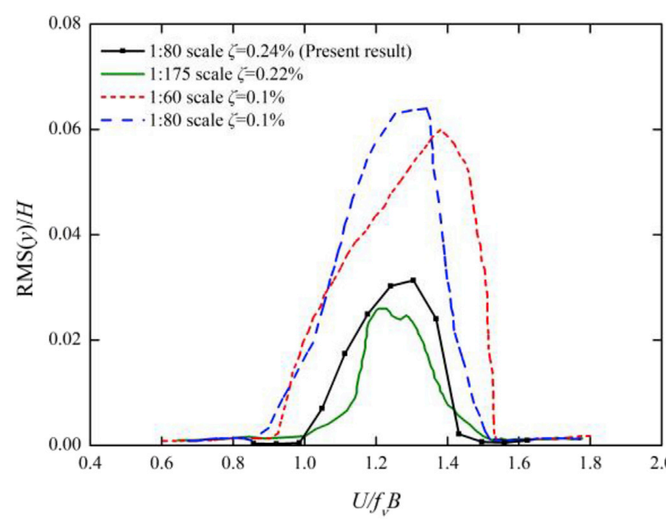


Fig. 8. V-VIV response comparison of the test models at different scales and damping ratios.

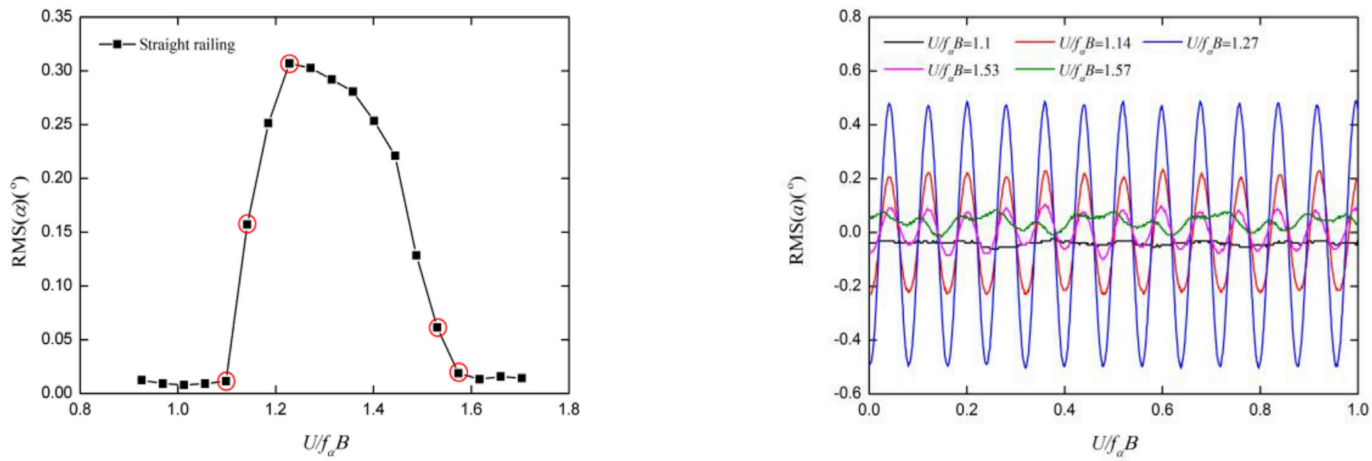
wind speed measured by the hot wire probe, f_ν and f_α are the vertical frequency (8.5 Hz) and torsional frequency (12.33 Hz) of the system respectively, B is the breath of the deck. The vertical vibration response, y, is given as the root mean square (RMS) vertical displacement normalized by the deck height (H). The torsional response, α, is given as the RMS torsional displacement. In order to quantify the control effectiveness to determine the optimal control parameters, the response reduction rate η was adopted, which was defined as follows:

$$\eta=-(P_c-P_0)/P_0$$

where P_0 is the RMS response peak of the model with traditional straight railings and P_c is the RMS response peak of the model with modified wavy railings. It's obvious that the larger η signifies the better control effects.

3.1. VIV response of the bridge deck with traditional railings

The vertical response versus reduced wind velocity ($U/f_\nu B$) for the base case is depicted in Fig. 7 (a), which shows the peak value of 0.031 is



(a) (b)

Fig. 9. (a) RMS of the torsional response against the reduced wind velocity; (b) Response time history.

Table 2
Test cases for investigating the influence of wavelength.

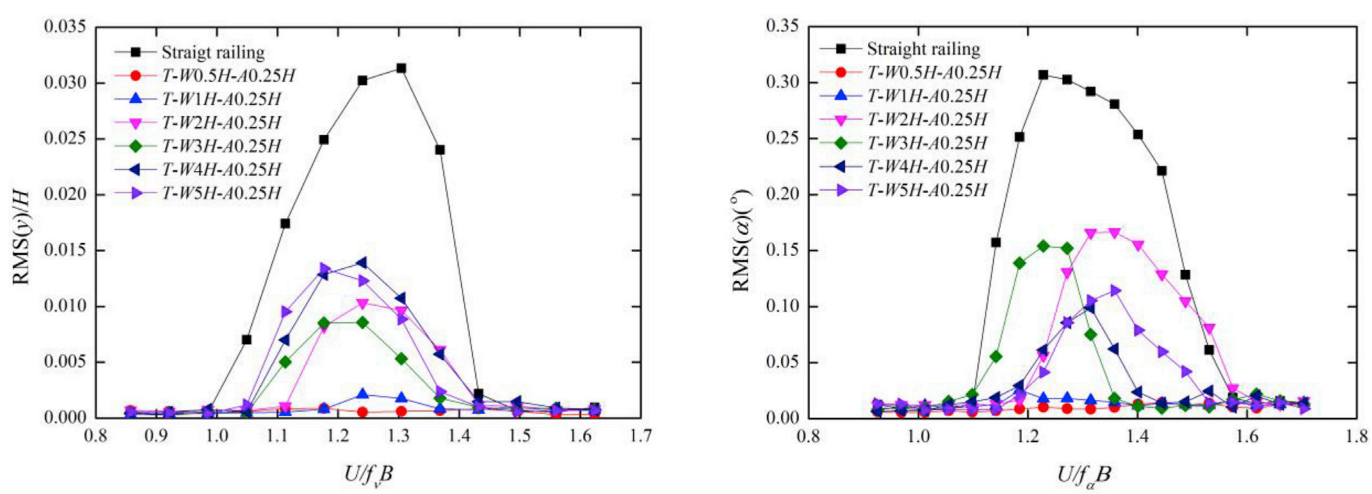
Case	Wavelength/H	Amplitude/H	Position of the wave
T-W0.5/1/2/3/4/5H-A0.25H	0.5/1/2/3/4/5	0.25	Outside the deck surface
T-W0.5/1/2/3/4/5H-A0.3H	0.5/1/2/3/4/5	0.3	Outside the deck surface
T-W0.5/1/2/3/4/5H-A0.36H	0.5/1/2/3/4/5	0.36	Outside the deck surface
T-W0.5/1/2/3/4/5H-A0.4H	0.5/1/2/3/4/5	0.4	Outside the deck surface

at $U/f_vB \approx 1.3$ within the previous identified range of $0.9 \le U/f_vB \le 1.5$ (Larsen and Wall, 2012). In Fig. 7 (b), the response time history extracted at 5 key point labeled in Fig. 7 (a) is presented. It can be seen that, during the test, $U/f_vB = 1.0$and $U/f_vB = 1.43$are respectively the lower and upper limit velocities of the vertical lock-in flow regime causing the harmonic oscillations. In the previous studies, the lock-in range obtained from a 1:80 section model was $1.0 \le U/f_vB \le 1.4$ (DMI, 1993), and the range of $1.1 \le U/f_vB \le 1.4$ was achieved in a 1:60 section model (Larose, 1992), which compare well with the present result of $1.0 \le U/f_vB \le 1.43$. Fig. 8 shows the comparison of RMS V-VIV amplitude between the present test and the previous section model tests at different model scales and damping ratios (Larsen and Wall, 2012). As expected, the amplitude of VIV decreases as the damping increases, and the present result ($\zeta = 0.24\%$) is smaller than the less damped 1:80 model in DMI (1993) ($\zeta = 0.1\%$), but close to the 1:175 model with the similar damping ratio ($\zeta = 0.2\%$). In addition, the torsional VIV was also observed in the test and the RMS torsional response versus reduced velocity ($U/f_\alpha B$) is plotted in Fig. 9 (a), which shows the torsional peak value of 0.31 is at $U/f_vB \approx 1.2$. Correspondingly, Fig. 9 (b) presents five responding time histories extracted from the key point marked in Fig. 9 (a), from which the torsional lock-in range of $1.1 \le U/f_vB \le 1.57$ is achieved.

3.2. Control effect of the wavy railing

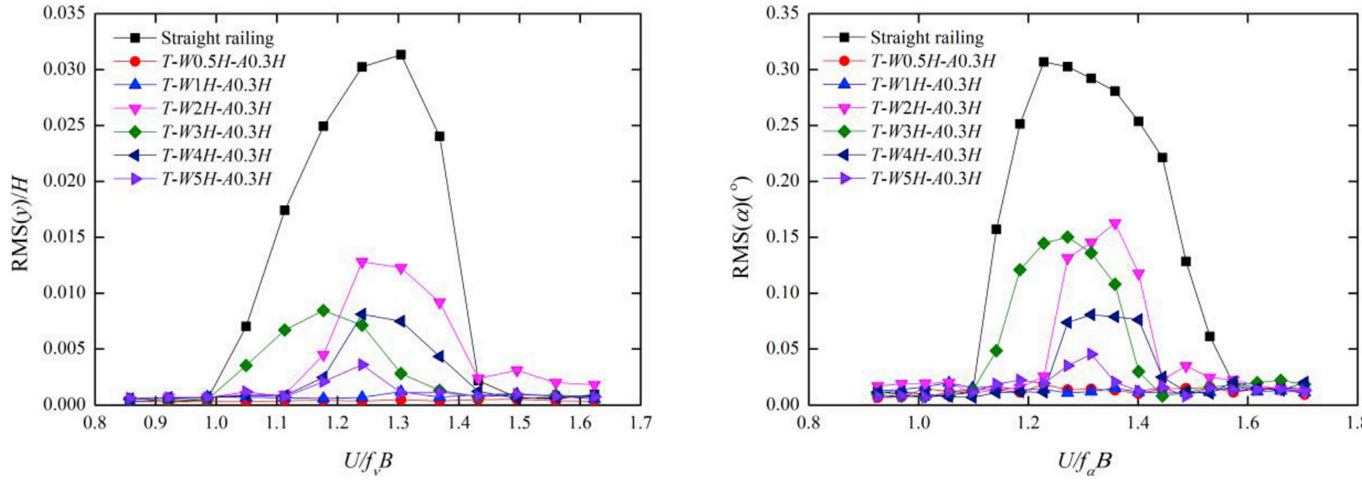
3.2.1. Influence of wavelength on control effect

In this section, for wavy railings, the effect of the wavelength was investigated under different amplitudes while fixing the installation position T (i.e. crossbars are located outside the deck surface). Six



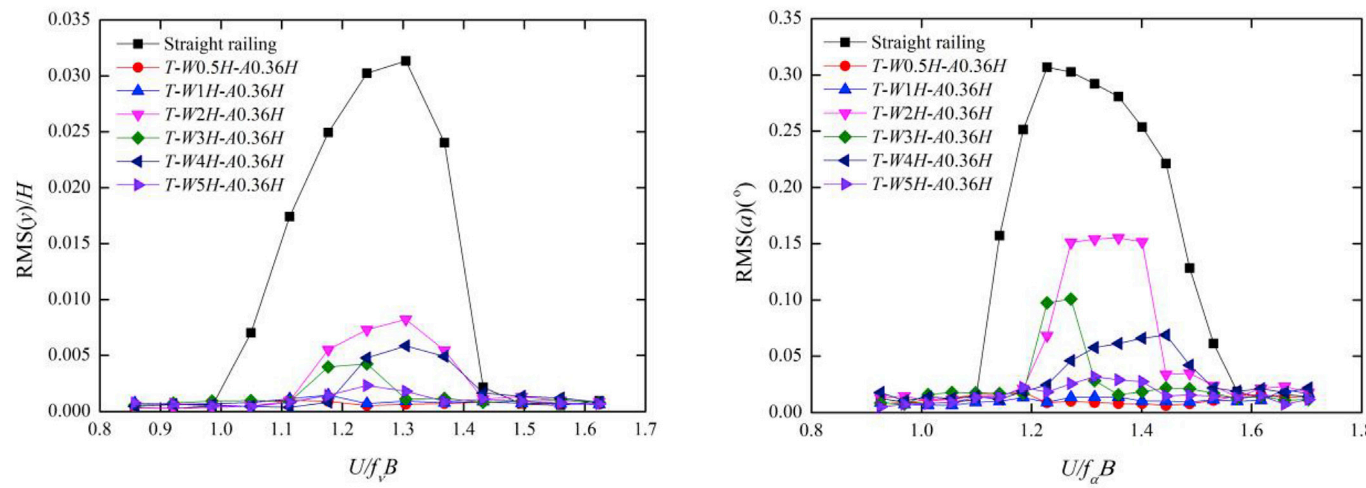
(a) (b)

Fig. 10. RMS of the (a) vertical and (b) torsional response with different wavelengths ($A/H = 0.25$).



(a) (b)

Fig. 11. RMS of the (a) vertical and (b) torsional response with different wavelengths ($A/H = 0.3$).



(a) (b)

Fig. 12. RMS of the (a) vertical and (b) torsional response with different wavelengths ($A/H = 0.36$).

wavelengths (i.e. $W/H = 0.5$, 1, 2, 3, 4 and 5) and four amplitudes (i.e. $A/H = 0.25$, 0.3, 0.36, and 0.4) were selected to constitute a total of 24 test cases, as listed in Table 2. Fig. 10 (a) and (b) respectively summarize the V-VIV and T-VIV response for values of $W/H = 0.5$, 1, 2, 3, 4 and 5 with the constant amplitude $A/H = 0.25$, and the corresponding results of the base case (Straight railing) are also incorporated for comparison. As can be seen, both the V-VIV and T-VIV can be completely suppressed in the case of low wavelengths, i.e. $W/H = 0.5$ and $W/H = 1$, and the maximum response reduction occurs for $W/H = 0.5$ yielding the V-VIV reduction of 97.2% and T-VIV reduction of 95.3%. In addition, the high values of wavelengths i.e. $W/H = 4$ and $W/H = 5$ exhibit the weakest control effect on the V-VIV, and the modest wavelengths i.e. $W/H = 2$ and $W/H = 3$ have the least effect on restricting the T-VIV. Considering the cases of $A/H = 0.3$ (see Fig. 11), the control effects of low wavelengths, especially $W/H = 0.5$, are still prominent, resulting in the substantial V-VIV reduction of 98.3% and T-VIV reduction of 93.9%. However, the cases with $W/H = 2$ and $W/H = 3$ present the poorest control performance for both the V-VIV and T-VIV. Similar conclusions can be obtained for amplitudes greater than 0.3H, i.e. $A/H = 0.36$ and $A/H = 0.4$ (see Fig. 12 and Fig. 13). Moreover, it's interesting to note that the cases named *T-W5H-A0.36H* and *T-W5H-A0.4H* also show a satisfactory control effect, which obtain the V-VIV reductions of 92.6% and 94.8%, the T-VIV reductions of 89.7% and 91.9%, respectively. This indicates that, within the studied range, when the amplitude increases to a certain extent, $W/H = 5$ can become another recommended wavelength in addition to $W/H = 0.5$ and $W/H = 1$. The phenomenon of available wavelengths appearing in two domains was also observed in (Lam and Lin, 2009). Fig. 14 summarizes the variations of the V-VIV and T-VIV response reduction rates with respect to the wavelength. As mentioned above, the significant response reduction of at least 90% can be achieved by all the cases with the wavelength of $W/H = 0.5$ or $W/H = 1$, indicating that the flow around the deck has an intrinsic preference for the disturbance of wavy railings with a wavelength of $0.5H$ and $1H$. Therefore, we can make the conjecture that the introduction of the critical spanwise disturbances of $0.5H$ and $1H$ alters the dynamics of the shear layers and prevents them from rolling up into spanwise vortices with a minimum amplitude. In addition, the curves of response reduction show a 'trough' character at $W = 2H$ except for $A/H = 0.25$, which illustrates that the formation of the vortex shedding is insensitive to the disturbance generated by the wavy railing with $W = 2H$, thus $2H$ is the unfavorable choice for wavelength

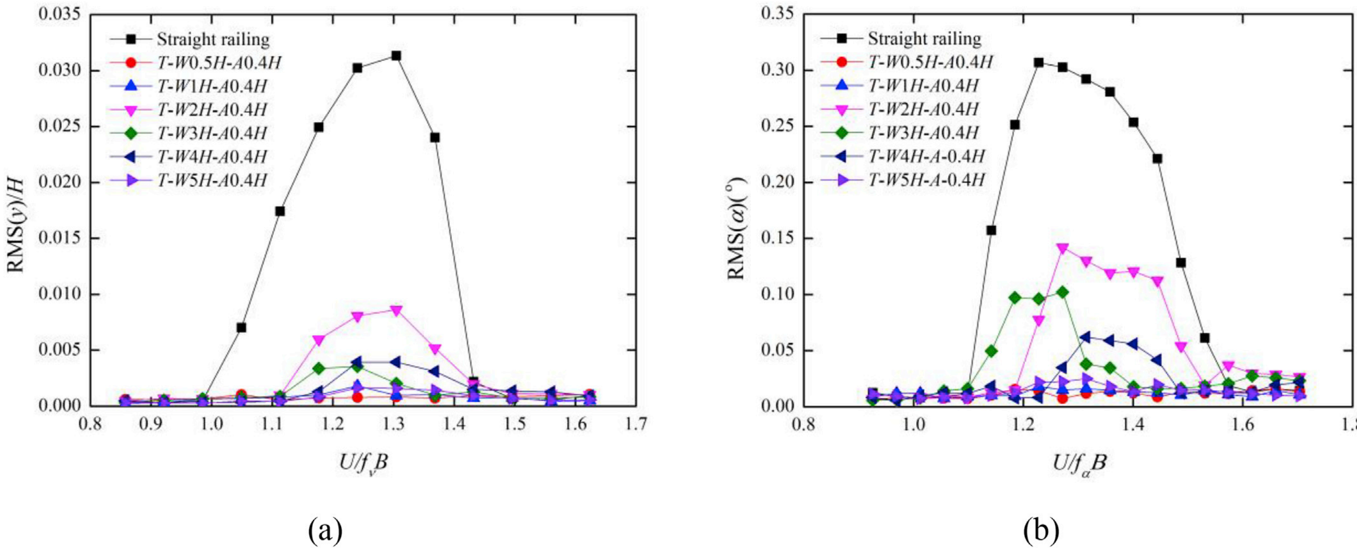


Fig. 13. RMS of the (a) vertical and (b) torsional response with different wavelengths ($A/H = 0.4$).

setting in most cases.

3.2.2. Influence of amplitude on control effect

To study the influence of the wavy railing's amplitude on the control effect, the selected cases in section 3.2.1 were regrouped according to the same wavelength (Table 3). Figss. 15–17 respectively show the results of the VIV respond versus reduced wind velocity for $A/H = 0.25$, 0.3, 0.36, and 0.4 with the constant wavelengths of $W/H = 0.5$, 1, 2, 3, 4 and 5, and the base case is also incorporated for comparison. As shown in Fig. 15, when the wavelengths are the optimal values i.e. $W/H = 0.5$ and $W/H = 1$, the remarkable control effect is insensitive to the amplitude selected in this research. Therefore, in this case, the minimum amplitude of $A/H = 0.25$ is the best choice due to the economic advantage. However, the observable 'amplitude dependence' characteristic exists in the cases where the VIV cannot be completely suppressed. Considering the cases with $W/H = 2$ shown in Fig. 16, larger amplitudes i.e. $A/H = 0.36$ and $A/H = 0.4$ have better control effects than smaller amplitudes $A/H = 0.25$ and $A/H = 0.3$ on controlling the V-VIV, and the maximum amplitude i.e. $A/H = 0.4$ also shows the best control effects on mitigating the T-VIV. A similar phenomenon is also observed when the wavelength is $W/H = 3$, and the increase of amplitude tends to narrow the lock-in range and reduce the response peak. When the wavelength increases further, the superiority of the large amplitude becomes more obvious. As can be seen from Fig. 17, when the wavelength is $W/H = 4$, with the amplitude

Table 3
Test cases for investigating the influence of amplitude.

Case	Amplitude/H	Wavelength/H	Position of the wave
T-W0.5H-A0.25/0.3/0.36/0.4H	0.25/0.3/0.36/0.4	0.5	Outside the deck surface
T-W1H-A0.25/0.3/0.36/0.4H	0.25/0.3/0.36/0.4	1	Outside the deck surface
T-W2H-A0.25/0.3/0.36/0.4H	0.25/0.3/0.36/0.4	2	Outside the deck surface
T-W3H-A0.25/0.3/0.36/0.4H	0.25/0.3/0.36/0.4	3	Outside the deck surface
T-W4H-A0.25/0.3/0.36/0.4H	0.25/0.3/0.36/0.4	4	Outside the deck surface
T-W5H-A0.25/0.3/0.36/0.4H	0.25/0.3/0.36/0.4	5	Outside the deck surface

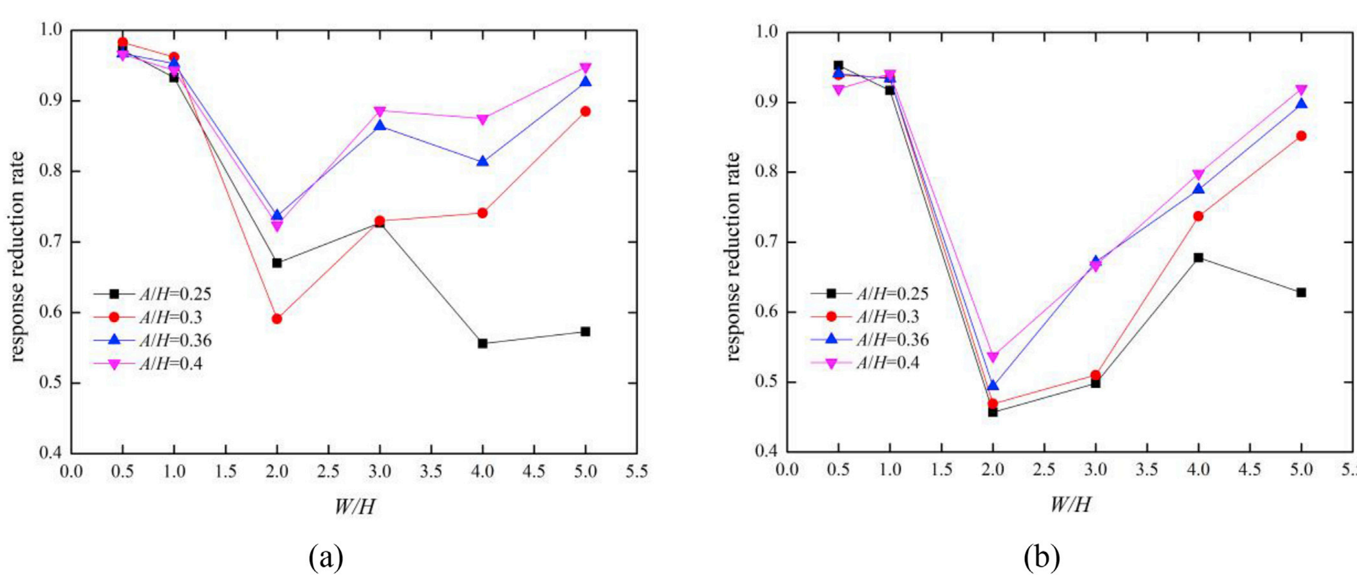


Fig. 14. (a) Vertical and (b) torsional response reduction rate versus wavelength.

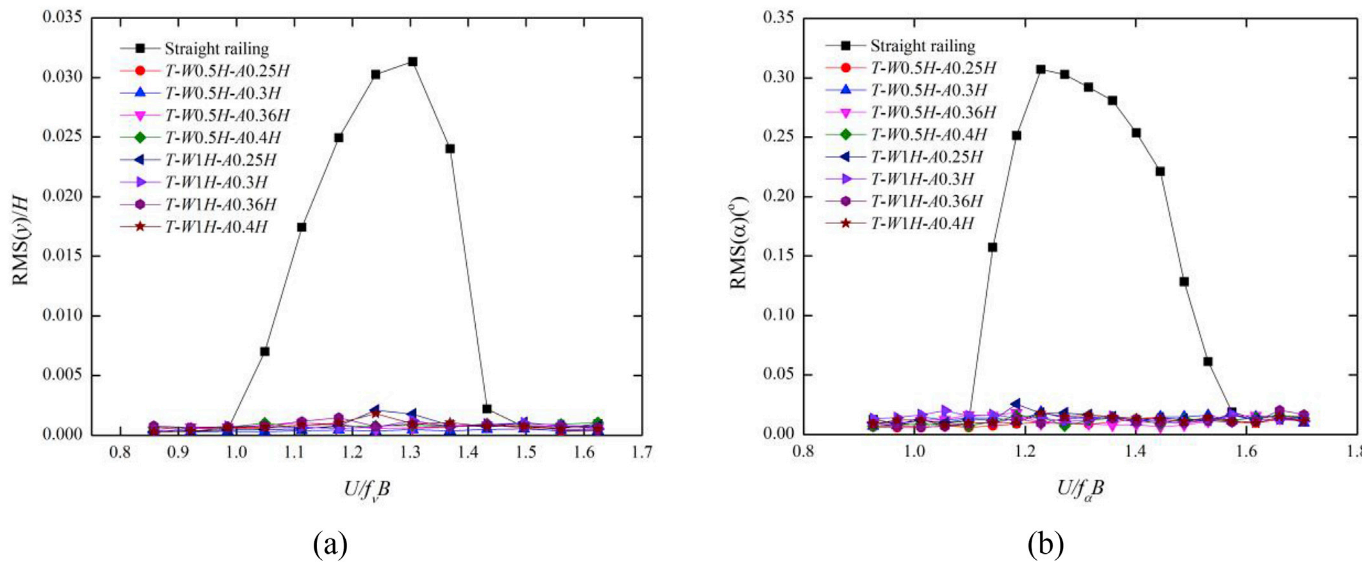


Fig. 15. RMS of the (a) vertical and (b) torsional response with different amplitudes ($W/H = 0.5$, 1).

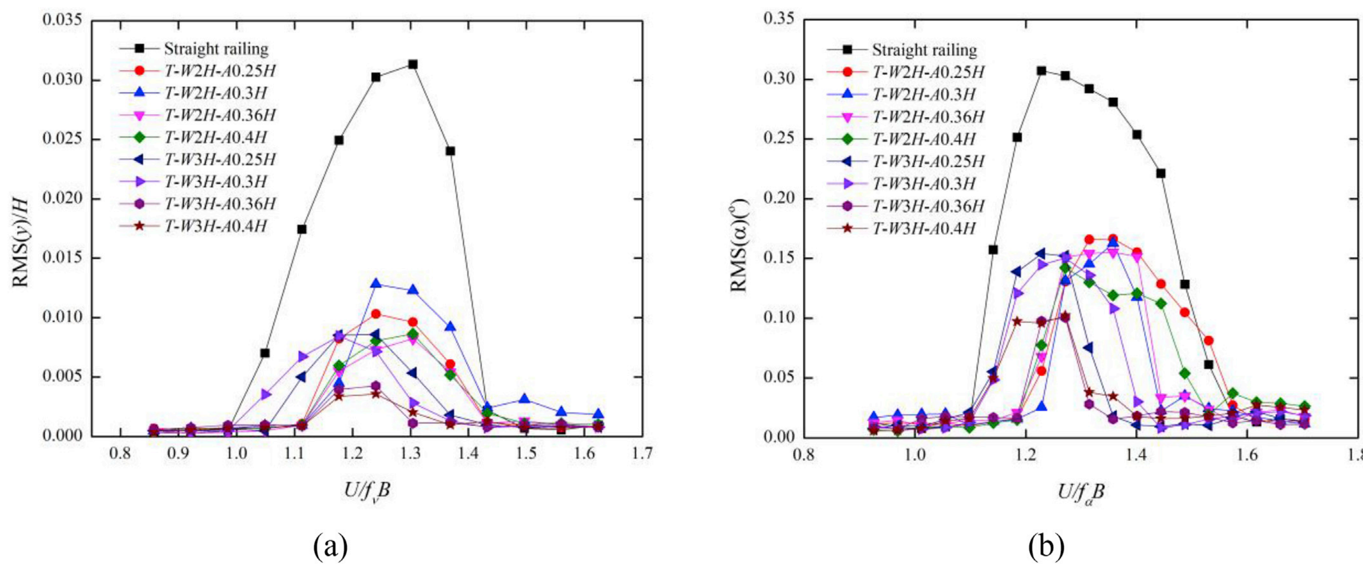


Fig. 16. RMS of the (a) vertical and (b) torsional response with different amplitudes ($W/H = 2$, 3).

increasing from $A/H = 0.25$ to $A/H = 0.4$, the V-VIV reduction increases from 55.6% to 87.5%, and the T-VIV reduction increases from 67.8% to 79.8%. Within the wavelength range investigated, $W/H = 5$ is the most sensitive value to amplitude. It can be found that with the amplitude increasing from $A/H = 0.25$ to $A/H = 0.4$, the promotion of V-VIV reduction rate of up to 37.5% is achieved, and the T-VIV reduction rate is increased by 29.1%. Especially when the amplitude increases to $A/H = 0.4$, the V-VIV and T-VIV are almost suppressed. Fig. 18 summarizes the variations of the V-VIV and T-VIV response reduction rates with respect to the amplitude. As can be seen, for the cases with the optimal wavelengths i.e. $W/H = 0.5$ and $W/H = 1$, the influence of the amplitude on control effects can be ignored, indicating that compared with amplitude, the appropriate setting of wavelength is the more dominant factor to achieve the satisfactory control effects. Regarding to the other cases i.e. $W/H = 2,3,4$ and 5, there is a general trend that the response reduction rate increases progressively with the increasing of amplitude, which illustrates that when the selected wavelength is not the best choice, a pronounced improvement of control effects can be obtained by increasing the amplitude. This phenomenon is quite similar to that observed in Bearman and Owen (1998), in which the straight bar was reshaped to the wave form for a better aerodynamic performance, and the wavy bar with larger amplitude exhibits more prominent superiority. According to Darekar and Sherwin (2001), the superiority of larger amplitudes is due to the increase of streamwise vorticity in the near wake that is diverted from the spanwise vorticity, and these three-dimensional redistribution of vorticity may effectively prevent periodic vortex shedding and hence suppress the VIV.

3.2.3. *Influence of installed position on control effect*

In this section, we designed 3 different positions to install the wavy railings, i.e. the wavy crossbars are completely outside the deck (*T*), half outside the deck (*M*) and completely inside the deck (*L*) (see Fig. 6) to investigate the optimal installation position. Based on the foregoing study, three representative wavy railings with the geometric parameters of ($W/H = 1$, $A/H = 0.36$), ($W/H = 3$, $A/H = 0.36$) and ($W/H = 5$, $A/H = 0.36$) were selected to constitute the total of 9 test cases as listed in Table 4. According to the test results, the control effects of wavy railings are highly sensitive to their installation positions. To be specific, for the

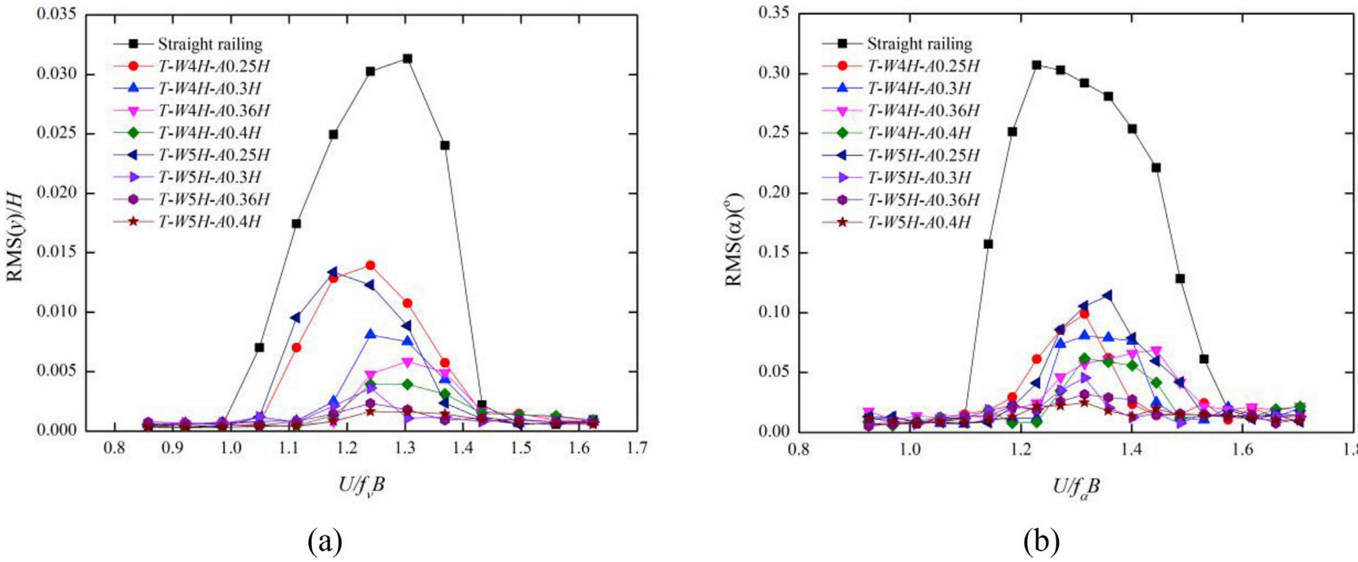


Fig. 17. RMS of the (a) vertical and (b) torsional response with different amplitudes (W/H = 4, 5).

cases with (W/H = 1, A/H = 0.36) shown in Fig. 19, when the wavy crossbars are located inside the deck surface (*L*), the V-VIV and T-VIV are mitigated only in a small degree. However, the control effect can be significantly improved by moving the wavy railings outward with half the peak-to-peak amplitude (*M*), resulting in the increase of V-VIV reduction of 54% and T-VIV reduction of 82.8%. And as the position is further moved outward until the waves are totally outside the deck (*T*), both the V-VIV and T-VIV can be completely suppressed. For the cases with (W/H = 3, A/H = 0.36) and (W/H = 5, A/H = 0.36), the similar influence rule of installation position can be also summarized in Fig. 19, which indicates that the best choice is to install the wavy railings in a position where the wavy crossbars completely outside the deck surface. Fig. 20 depicts the variation of the V-VIV and T-VIV response reduction rates with respect to the installation position. The variation trend for either V-VIV or T-VIV is that the response reduction rate gradually increases with the outward moving of the installation position. This may stem from the fact that during the course of transit to the wake, the disturbance provided inside the deck is prone to be attenuated by the interaction with the deck surface. However, since the process of the disturbance shifting is not affected by the deck, the waves totally outside the deck surface can generate stronger disturbances, which act more directly on the wake and thus lead to better control effects.

Table 4
Test cases for investigating the influence of installation position.

Case	Position of the wave	Wavelength/*H*	Amplitude/*H*
T-W1H-A0.36H	Outside the deck surface	1	0.36
M-W1H-A0.36H	Half outside the deck surface	1	0.36
L-W1H-A0.36H	Inside the deck surface	1	0.36
T-W3H-A0.36H	Outside the deck surface	3	0.36
M-W3H-A0.36H	Half outside the deck surface	3	0.36
L-W3H-A0.36H	Inside the deck surface	3	0.36
T-W5H-A0.36H	Outside the deck surface	5	0.36
M-W5H-A0.36H	Half outside the deck surface	5	0.36
L-W5H-A0.36H	Inside the deck surface	5	0.36

3.2.4. Control effects influenced by the wind attack angles

The control effects of the wavy railings on mitigating VIVs under the wind attack angle (WAA) of ±3°are shown in Fig. 21, which include the

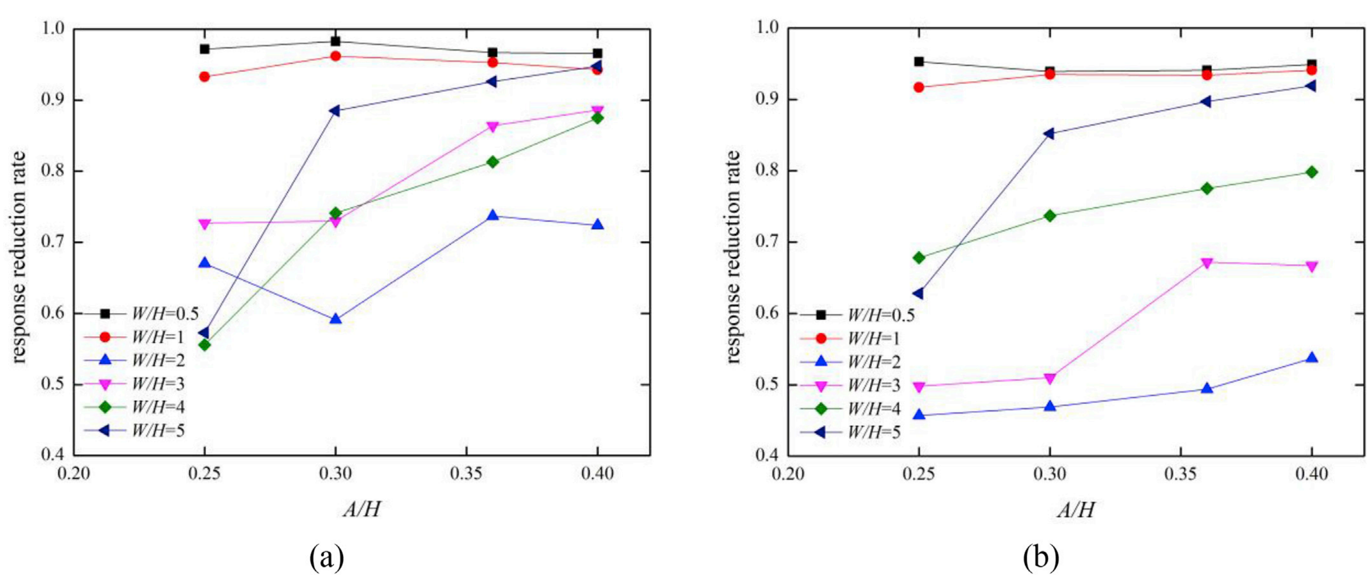


Fig. 18. (a) Vertical and (b) torsional response reduction rate versus amplitude.

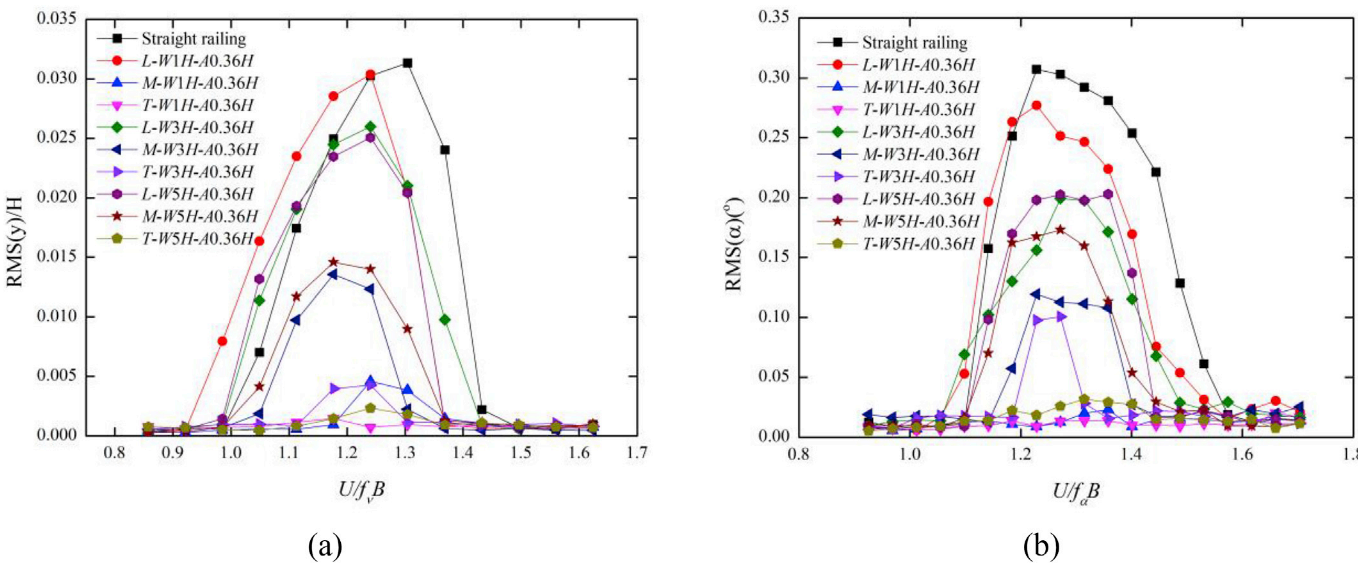


Fig. 19. RMS of the (a) vertical and (b) torsional response in different positions ($W/H = 1, 3, 5$; $A/H = 0.36$).

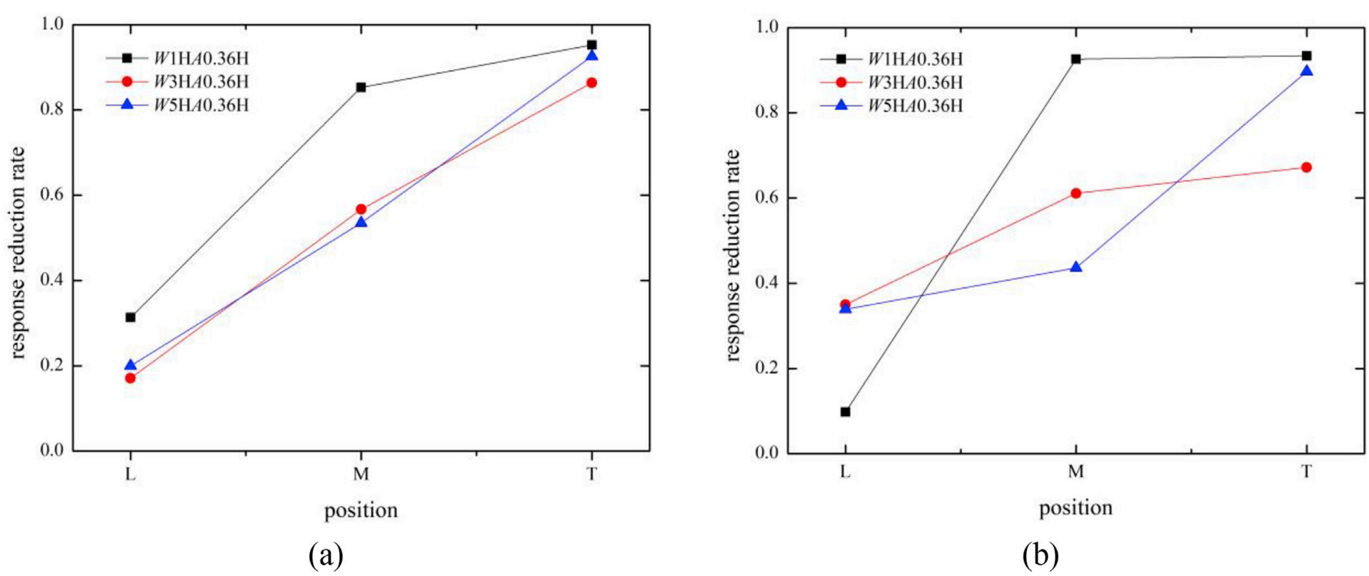


Fig. 20. (a) Vertical and (b) torsional response reduction rate versus position.

plots of the RMS of V-VIV and T-VIV as function of the reduced wind speed for three cases with remarkable control effects under the WAA of 0°named *T-W*0.5*H-A*0.3*H*, *T-W*1*H-A*0.4*H* and *T-W*5*H-A*0.4*H*. Considering the base case under the WAA of + 3°, the dimensionless RMS peak values of V-VIV and T-VIV are 0.027 and 0.244, respectively, which decreased slightly compared with those under the WAA of 0°, which are 0.031 and 0.307. The control effects of *T-W*0.5*H-A*0.3*H* and *T-W*1*H-A*0.4*H* under the WAA of + 3°are as significant as under the WAA of 0°, both the V-VIV and T-VIV are completely suppressed. Although *T-W*5*H-A*0.4*H* exhibit a little weaker control effect for the V-VIV, the onset wind velocity is delayed, the lock-in range is narrowed and the vertical response is greatly reduced. In addition, the T-VIV can be completely suppressed by *T-W*5*H-A*0.4*H*. For the base case under the WAA of − 3°, the dimensionless RMS peak values of the vertical response is 0.041, which is larger than that at the WAA of 0°, however, there's no T-VIV occurs at the WAA of − 3°. It can be seen that all the selected control cases i.e. *T-W*0.5*H-A*0.3*H*, *T-W*1*H-A*0.4*H* and *T-W*5*H-A*0.4*H* show substantial control capacity for the V-VIV and do not deteriorate the T-VIV performance at the same time.

4. Conclusions

Wind tunnel experiments were carried out to verify the control effect of the proposed wavy railings on the bridge VIV. The influence of self-defined control parameters (i.e. wavelength *W*, peak-to-peak amplitude *A* and installation position) were studied in detail. The major conclusions are as follows:

The VIV performance of the bridge-deck model is significantly improved by modifying the straight railing to the wavy railing. Compared to the straight, non-wavy railing, 98.3% V-VIV reduction and 95.3% T-VIV reduction are obtained for the optimized configuration of the wavy railing.

The wavelength is a more dominant geometric parameter for controlling VIV compared to the amplitude, and the flow around the bridge in this study has an intrinsic preference for the disturbance of particular wavelengths of 0.5*H* and 1*H*. Considering the wavy railings without the optimal wavelength, the pronounced improvement of control effects can be achieved by increasing the value of the amplitude, especially for the cases with $W/H = 5$, the increase of 37.5% V-VIV reduction and 29.1% T-

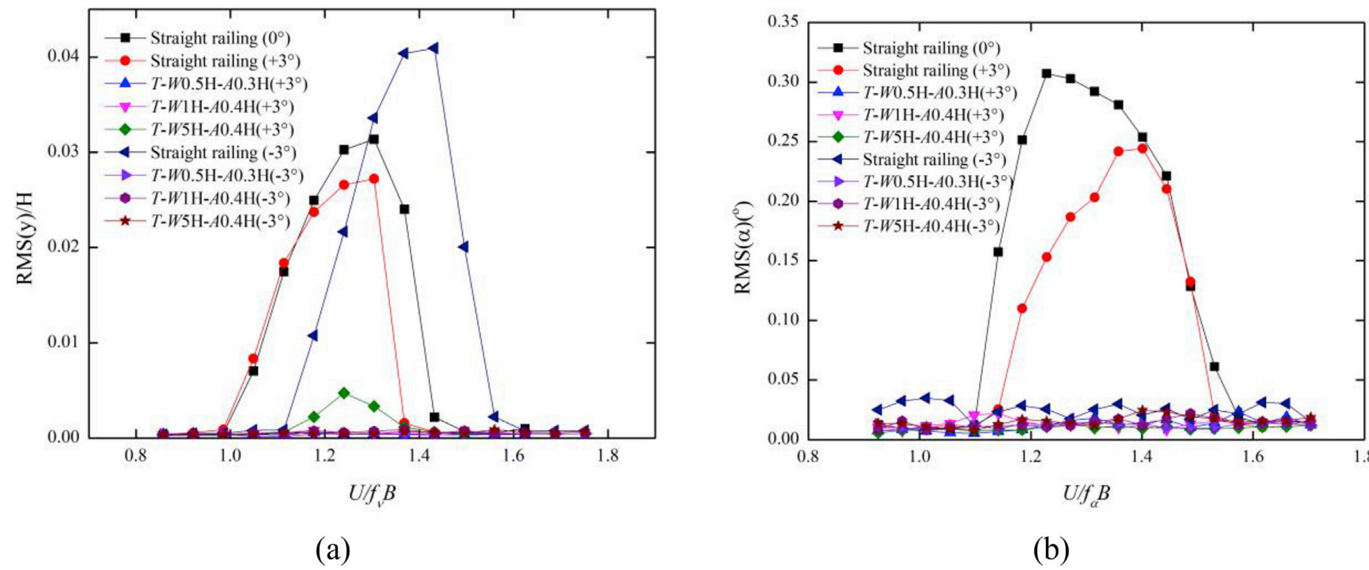


Fig. 21. RMS of the (a) vertical and (b) torsional response under the WAA of +3°and −3°.

VIV reduction are achieved respectively by increasing the amplitude from $A/H = 0.25$ to $A/H = 0.4$.

The control effect gradually increases as the wavy railings are moved outward, thus the installation position where the wavy crossbars are totally outside the deck edge is the best choice. In addition, both V-VIV and T-VIV can be completely suppressed by the wavy railings with the optimal wavelength of $W/H = 0.5$ under the wind attack angle of $\pm 3^{\circ}$.

As mentioned above, the wavy railing with optimized parameters can effectively suppress the VIV of the bridge studied in this paper. However, their applicability to other cross-sections of bridges, and their influence on the flutter property and aerodynamic forces of the deck need to be further investigated.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jian Zhan: Methodology, Validation, Writing - original draft, Writing - review & editing, Data curation. **Dabo Xin:** Conceptualization, Resources, Visualization, Project administration, Funding acquisition. **Jinping Ou:** Conceptualization, Supervision. **Zhiwen Liu:** Methodology, Formal analysis, Investigation.

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